

security design plays a vital role. This is done at four levels. Infrastructure security manages security at the physical device level using device controllers, edge, gateway and cloud. Platform security manages security at runtime in edge and gateway to handle access control and protect data privacy requirements. Network security manages and monitors data encryption during network transfers — a two-way handshake ensures secured clients are connected, and security logs are audited for anomalous behaviour, if any. Application security manages native and third-party services handling data, and helps protect IoT services from cyberattacks.

Potential challenges in IoT security include data privacy compromised when using third-party services, unprotected data storage, lack of role

based access control (RBAC), weak cloud architecture to protect public services, and insecure or outdated IoT device components being exposed to cyberattacks.

Guidelines for developing a security strategy for IoT applications

Some of the core principles for developing a security strategy for IoT applications and infrastructure services are:

- Data encryption at rest to protect data when it is stored in the data storage or application storage facility. This could be through encryption or device level cipher protection.
- Data encryption in transit or in motion to protect data when transporting it through a communication protocol. This could

be through a secured tunnel or safeguarding network services using a secured protocol like SSL, TLS, FTPS, etc.

- Device identity to protect unauthorised devices being connected to the IoT network of services — each device must have a unique identity. This will be used to authenticate (verify device access) and authorise (permit device connection) in the IoT ecosystem of connected devices.
- Using a token-based authentication and authorisation framework like OAuth 2.0 to use API based services for connecting and communicating with IoT applications.
- Policy management and role-based access control to enable resource level access management for all the connected services.

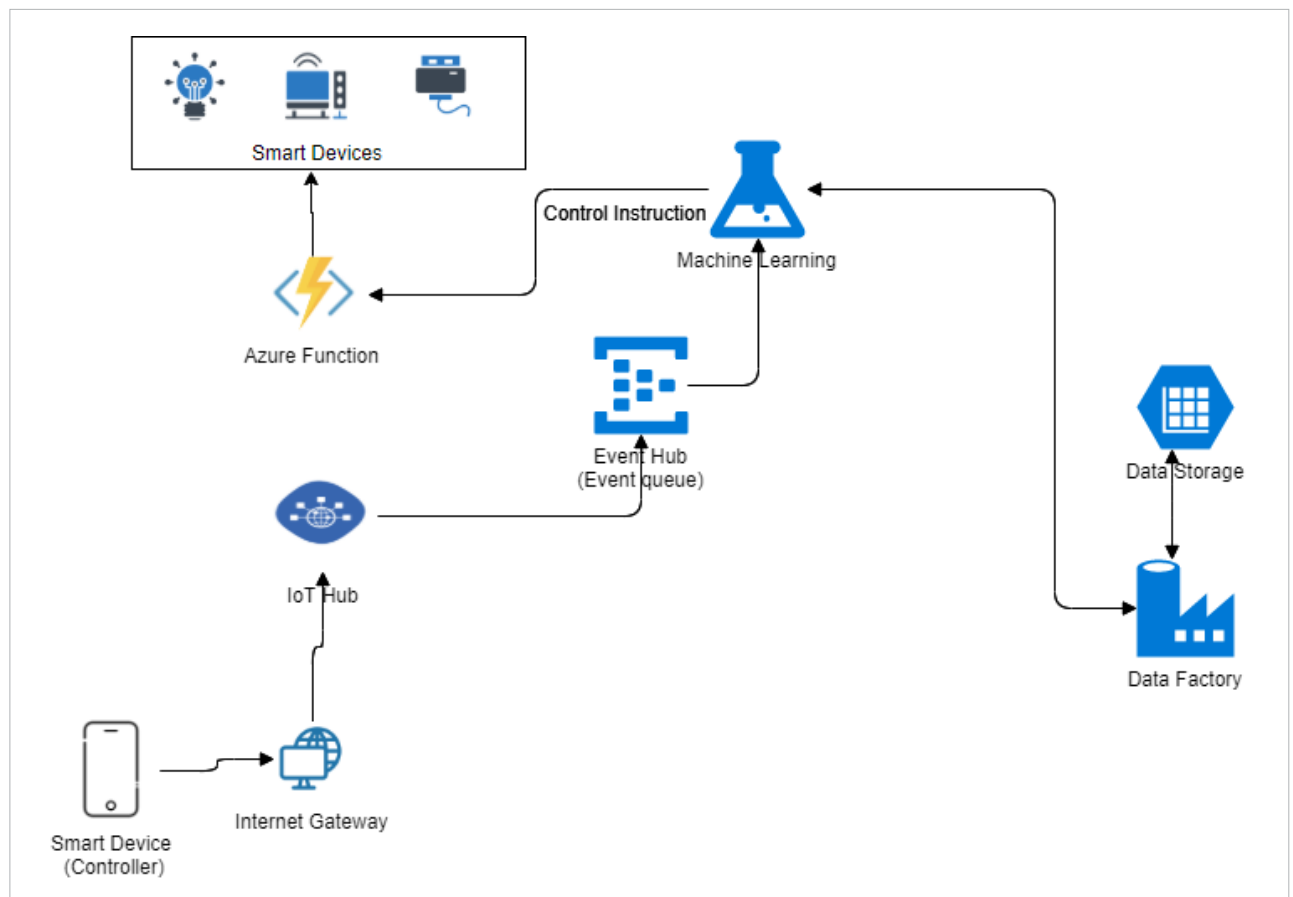


Figure 2: Smart home reference architecture with Azure IoT services